

The Optimal Endpoint Hardy Error under a Spectral Mass Budget

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Abstract

We solve the endpoint approximation problem inside the genuine finite normal spectral class. If all eigenvalues lie in $|\mu| \leq q$ and their squared mass is at most M , the least possible endpoint H^2 mismatch has squared norm asymptotic to $d/\log M$, where $d = \log(q_*/q)$. The lower bound uses the all-order deterministic coefficient law; the upper bound uses exact integer spectral prefixes. The theorem gives a sharp information-class constant but does not prove that the actual noisy family converges.

1 Extremal spectral error

For a finite conjugate multiset $\mathcal{S} \subset \{|\mu| \leq q\}$, put

$$\tau_n(\mathcal{S}) = \sum_{\mu \in \mathcal{S}} \mu^n, \quad M_2(\mathcal{S}) = \sum_{\mu \in \mathcal{S}} |\mu|^2.$$

Define

$$E_{\text{spec}}(M) = \inf_{M_2(\mathcal{S}) \leq M} \left(\sum_{n \geq 2} \frac{|\tau_n(\mathcal{S}) - a_n|^2 \rho_*^{2n}}{n^2} \right)^{1/2}.$$

Let $s = q_*/q$ and $d = \log s$.

Theorem 1.1 (sharp endpoint mass law). *As $M \rightarrow \infty$,*

$$\boxed{E_{\text{spec}}(M)^2 \sim \frac{d}{\log M}.}$$

Proof. For every admissible spectrum, $|\tau_n| \leq Mq^{n-2}$. Fix $\varepsilon > 0$ and put

$$n_M = \left\lceil \frac{(1 + \varepsilon) \log M}{d} \right\rceil.$$

Uniformly for $n \geq n_M$,

$$\frac{|\tau_n|}{q_*^n} \leq q^{-2} M s^{-n} \leq q^{-2} M^{-\varepsilon} \rightarrow 0,$$

whereas RH-268 gives $a_n/q_*^n \rightarrow 1$. Therefore

$$E_{\text{spec}}(M)^2 \geq (1 - o(1)) \sum_{n \geq n_M} \frac{1}{n^2},$$

and hence

$$\liminf_{M \rightarrow \infty} \log M E_{\text{spec}}(M)^2 \geq \frac{d}{1 + \varepsilon}.$$

Letting $\varepsilon \downarrow 0$ proves the lower bound.

Conversely, RH-316 gives a spectrum matching the first N anchors exactly, and RH-317 bounds its mass by Cs^N . Its endpoint-scaled spectral tail satisfies

$$\sum_{n>N} \frac{|\tau_n|^2 \rho_*^{2n}}{n^2} \leq C' \sum_{n>N} \frac{s^{2(N-n)}}{n^2} = O(N^{-2}),$$

whereas the target tail has squared norm

$$\sum_{n>N} \frac{|a_n|^2 \rho_*^{2n}}{n^2} = N^{-1}(1 + o(1)).$$

Thus the mismatch squared norm is $N^{-1}(1 + o(1))$. For a budget M , choose the largest N with $Cs^N \leq M$; then $N = (\log M)/d + O(1)$, proving the upper bound and the asymptotic. $\square$

Numerically,

$$d = 0.3377217782684642\dots, \quad \sqrt{d} = 0.5811383469264992\dots$$

Corollary 1.2 (actual-family universal lower-rate bound). *If the actual complement has squared mass $M_\sigma \leq \sigma^{-1}$, then*

$$\liminf_{\sigma \downarrow 0} \log(1/\sigma) \|g_\sigma\|_{H^2(\rho_*)}^2 \geq d.$$

Proof. Repeat the lower-bound argument with $n_\sigma = \lceil (1 + \varepsilon) \log(1/\sigma)/d \rceil$. The actual mass cap gives, uniformly for $n \geq n_\sigma$,

$$\frac{|\tau_{\sigma,n}|}{q_*^n} \leq q^{-2} \sigma^{-1} s^{-n} \leq q^{-2} \sigma^\varepsilon \rightarrow 0.$$

The same reciprocal-square tail yields $d/(1 + \varepsilon)$, and then $\varepsilon \downarrow 0$ gives the claim. This direct proof applies to the countable Hilbert–Schmidt complement spectrum as well as to finite spectra. $\square$

Remark 1.3. The lower bound tends to zero and is compatible with both convergence and nonconvergence. The constructive upper spectra are not identified with the actual noisy complement. Gates A–E remain false/open.